‘BUDGET APP’ EVOLUTION

This document is a continuation of Assignment 1: Sketching User Experience, except (hopefully) much easier to digest

What is covered

- I **identified issues** by reviewing the recordings of user testing my paper prototype
- I built an **experience map** by recording additional information about how a wider audience might complete the same task, and documented **opportunities for improvement**
- I used these to **iterate** on my design, by **annotating** issues, and rebuilding the wire flows from the ground up, based on the needs of the users.
- I documented **patterns**, **organizational and navigational models**, and how I facilitated **information seeking behaviors** present in my wire flows
- I created a **high-fidelity** black-and-white **prototype** in Figma, based on my improved wire flows.
- I performed additional **testing** on one external participant

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1. Discovery

Before redesigning my app, I need to learn how a broader audience saves.

Folder - https://rmiteduau-my.sharepoint.com/:f:/g/personal/s3972407_student_rmit_edu_au/EhKoO96W3GZDu_-ywulyE1oBI2gF7kLYr2Q155YsyHna_w?e=EyaUVj

Figma - https://www.figma.com/design/MroQlMv2F2rH9TdkELAAbF/Savings-App?node-id=114-846&t=WOaKoleNjYp14Bg8-1

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Prototype testing video - https://rmiteduau-my.sharepoint.com/:v:/g/personal/s3972407_student_rmit_edu_au/ERif4QC70ANBt8NfRJ23cjABR8IVA-SKAipOl4bTOpOlsA?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=8onS5B

Interview 1

Audio M4A

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Screen Recording

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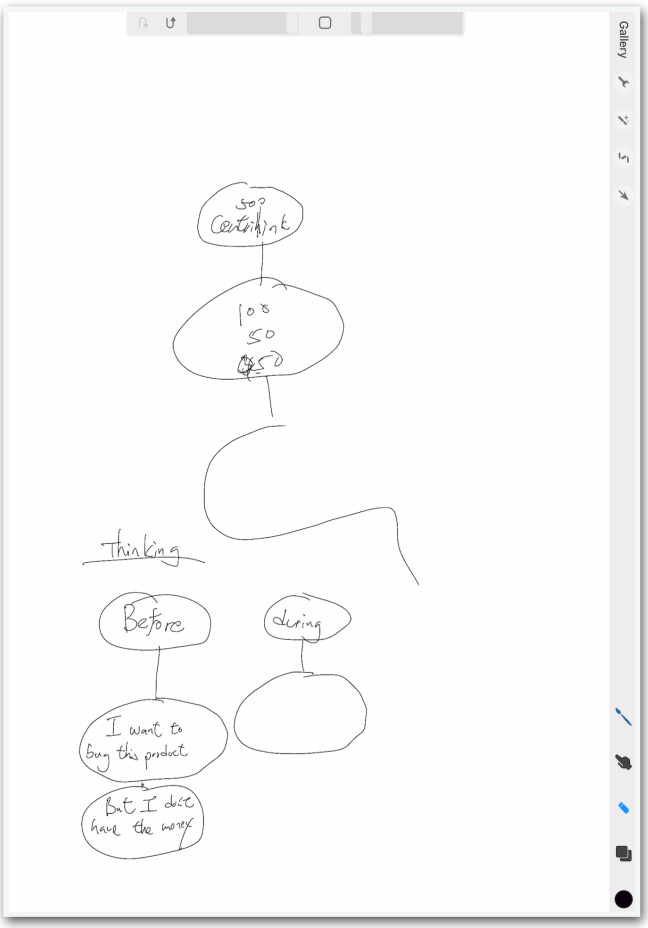
Mind Map



Questions and Answers

Task

Please describe how you would typically go about saving money?

Generally has no formal system for saving money, they tent to spend as much as needed without actively saving. when they do save, its for specific goals, like buying tech products. They have tried using tools for managing their expenses but found them too complicated and messy. they prefer something simple to categorize their spending.

Goals & Challenges

What goals do you have in mind when you are saving money?

Primary goals are to aim to save as much as possible, but they don't save regularly.

Have you saved for anything in particular?

Just tech related items like phones, speakers, etc.

Which parts of saving if any do you have difficulty with or find frustrating?

they find it difficult to calculate and log expenses, which leads to frustration. their tendency to spend freely also poses a challenge to reaching their savings goals.

Feelings

How are your feeling before you start a saving goal?

Frustrated because they don't have enough money and believe they are bad at saving.

What are your feelings during?

They feel motivated to save for the item they set out for

What do you feel after?

A feeling of accomplishment and happiness now that they have the item and they can look back and see their success.

how do you think your feelings influence how you save?

They are heavily influenced by motivation and desire.

Thoughts

What are you thinking about before you save for something?

Before starting to save, P1 considers their current money or lack of money and how long it might take to save

during saving

They don't think they are good at saving which leads to feelings of frustration

after saving

They look back and think about how they stuck to their plan and now have the item they wanted

Context

Where would you normally be when your saving for something? (Location)

At home, during a commute. Generally in their free time.

How long does it normally take to start a saving goal?

It typically takes them a few days to plan a new savings goal

How often would you do this task?

Typically on approximately a monthly basis.

actors

Simple systems for categorizing expenses

Product they want to save up for

artefacts

The product they are saving for

The phone or computer they use to mange their spending

Interview 2

Audio M4A

https://rmiteduau-my.sharepoint.com/:u:/g/personal/s3972407_student_rmit_edu_au/EfGH14i-Y1JJhKChC-MM2ksB3i6ERt2DkHlpSpc5g9chVQ?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=Nb0tq9

Screen Recording

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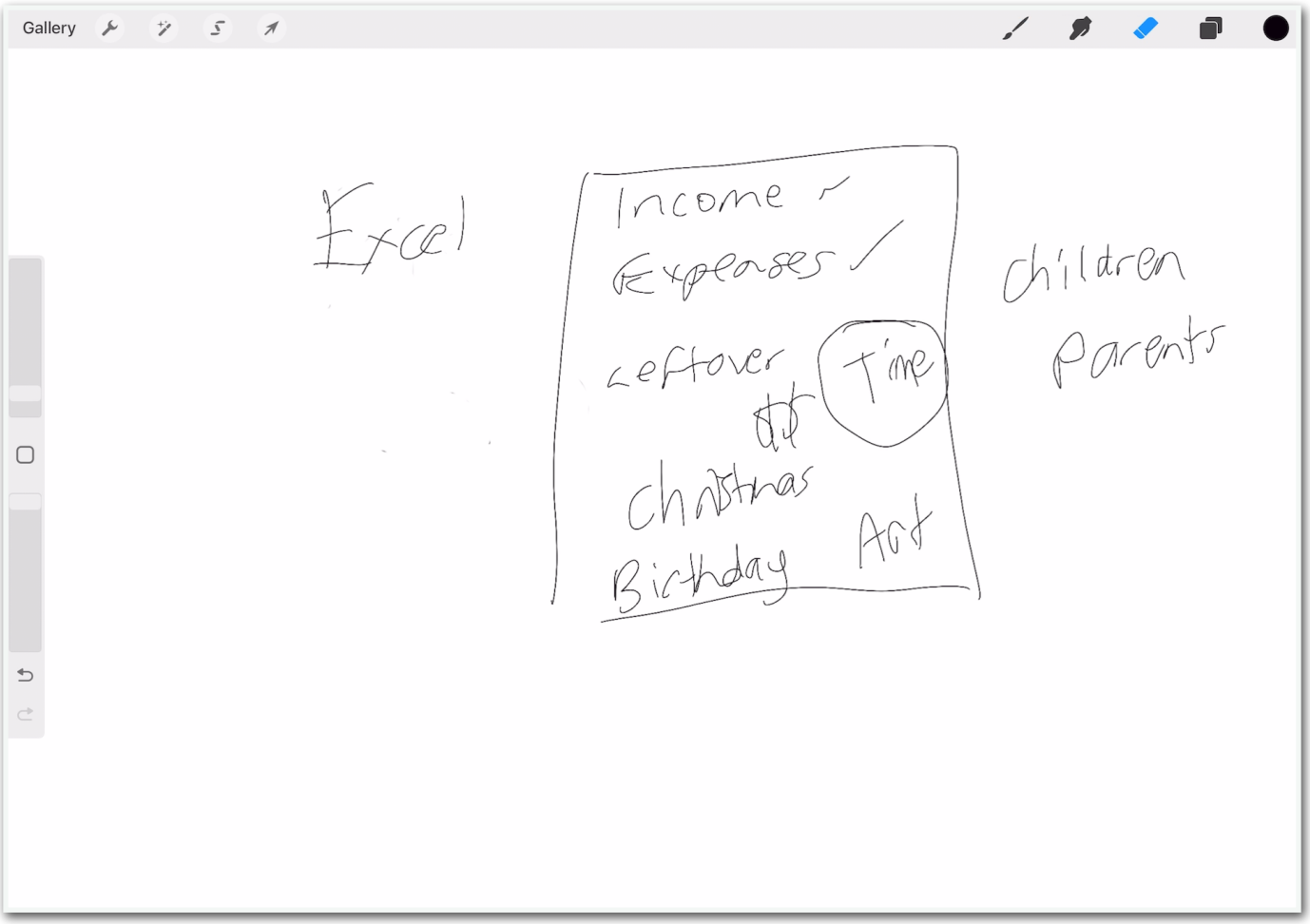
Mind Map



Questions and Answers

Task

Please describe how you would typically go about saving money?

They use an excel spreadsheet to track their income and expenses to calculate their leftover money. once they know the spare amount they have, They prioritize savings based on what they need (birthdays, Christmas, etc).

Goals & Challenges

What goals do you have in mind when you are saving money?

Typically their goal is based in needs or necessity, eg. events or birthdays.

Have you saved for anything in particular?

Examples they listed, like vacuum cleaners, tend to be more out of necessity

Which parts of saving if any do you have difficulty with or find frustrating?

They find it hard to prioritize saving over other distracting expenses.

Feelings

How are your feeling before you start a saving goal?

They feel excited about the idea of reaching a goal

What are your feelings during?

their feelings depend on their progress towards their goal, if they are behind they might feel anxious, however they tend to feel great when they are on track.

What do you feel after?

They feel great about reaching their goal

how do you think your feelings influence how you save?

They do not believe they are influenced by their feelings.

Thoughts

What are you thinking about before you save for something?

They think practically, they consider their income and expenses and the cost vs benefit of their goal.

during saving

re-evaluate their priorities and ensure the goal is still necessary. they also adjust their income and expenses as they change.

after saving

They start thinking about the next thing they need to save for

Context

Where would you normally be when your saving for something? (Location)

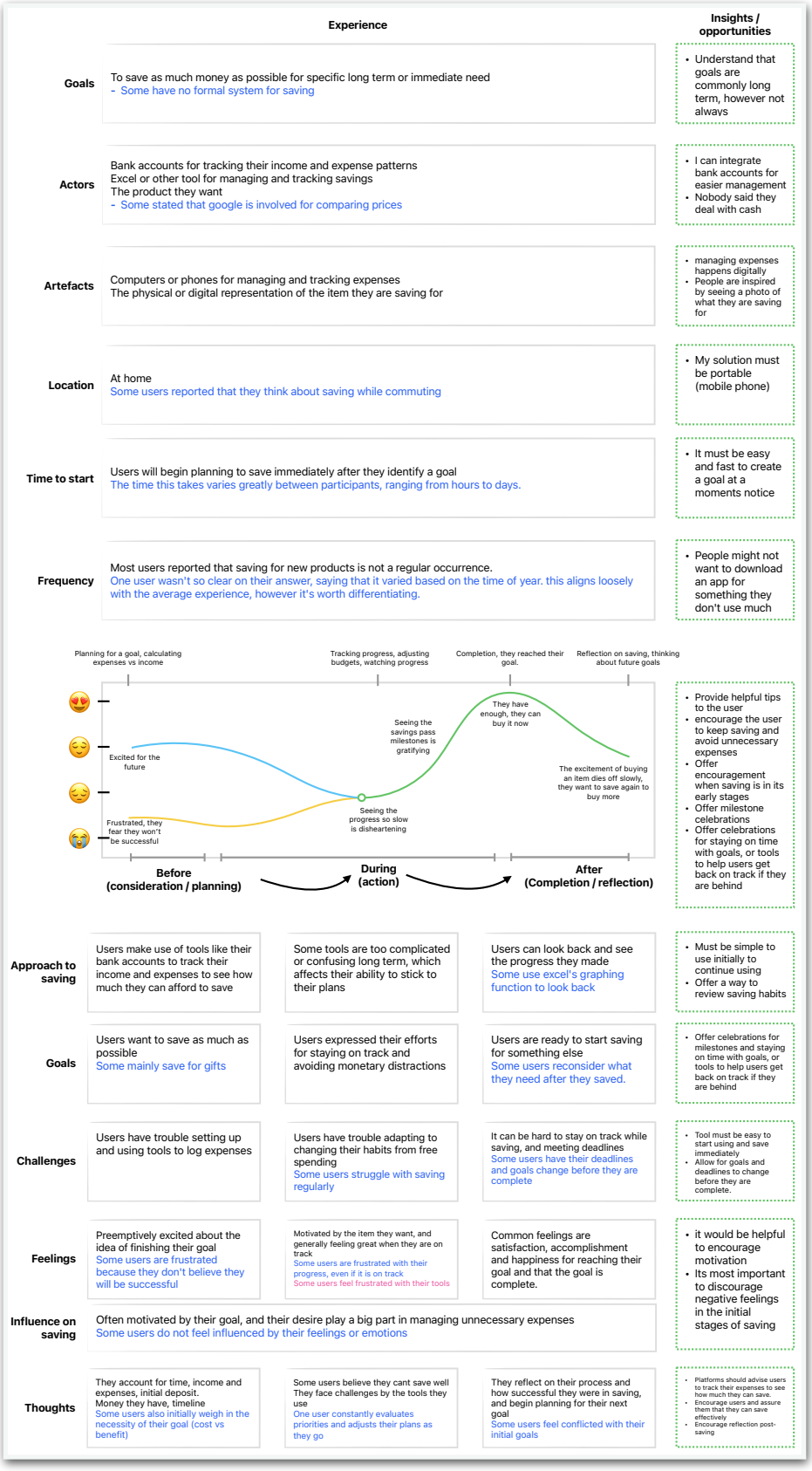
They typically plan and manage savings at home.

How long does it normally take to start a saving goal?

They start saving immediately after identifying a goal
How often would you do this task?
how often they save depends on the time of year

actors
The thing they are saving for
Excel to manage savings
artefacts
The thing they are saving for
excel spreadsheets

Experience Map



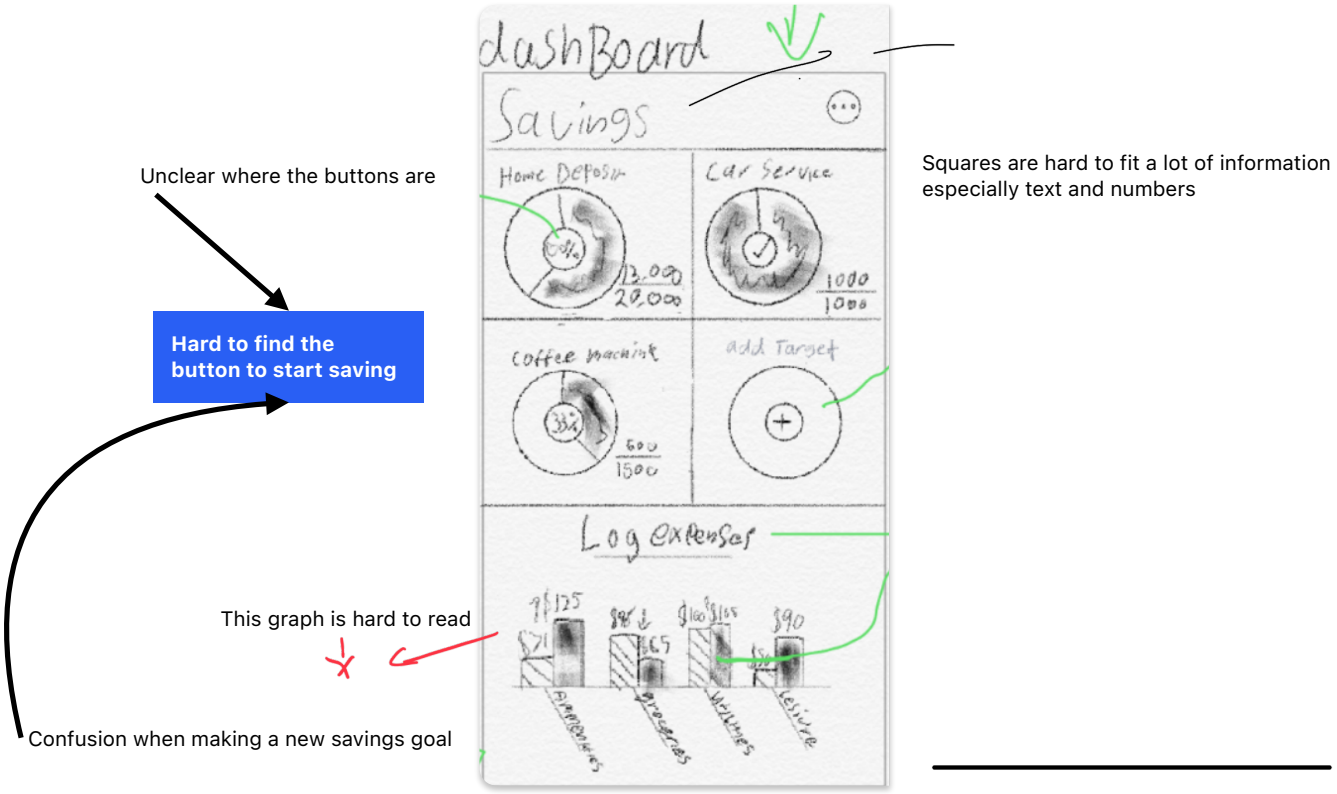
2. Iteration

Issues with current designs.

Looking back at my old designs and testing from A1 Paper prototypes, I asked myself questions like what went wrong when testing? What didn't work? or what didn't the users like?

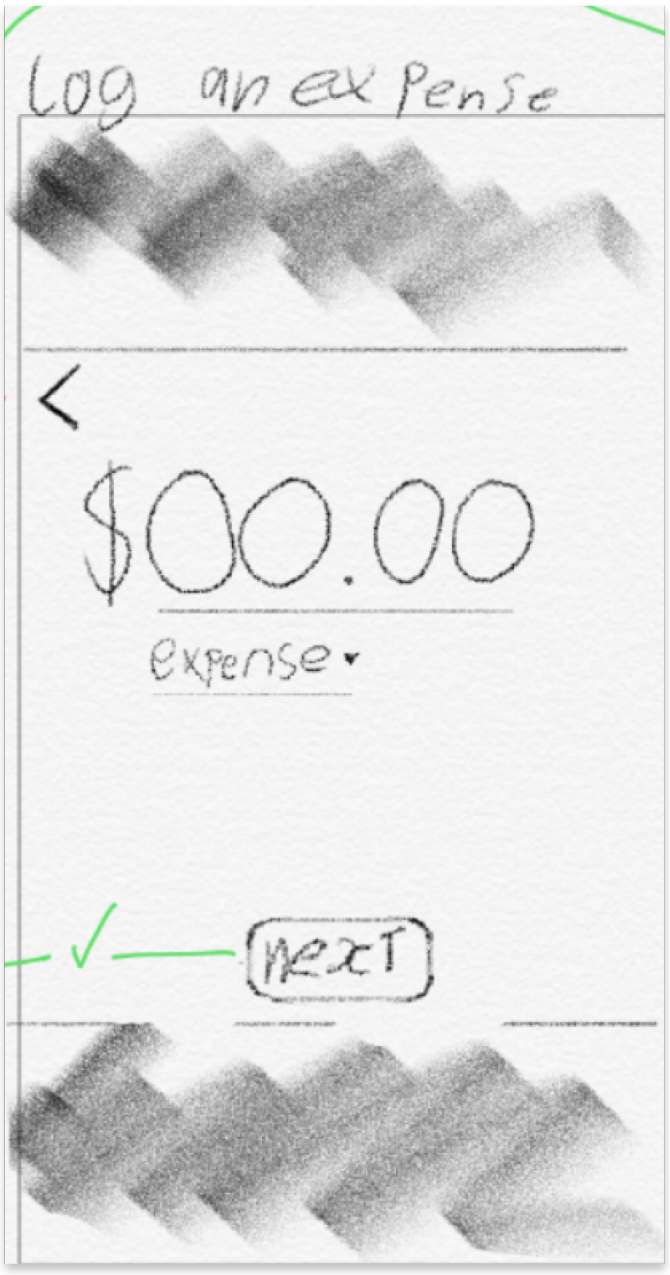
Dashboard

Users I tested had a lot of difficulty finding the appropriate buttons to press for the task they had, as well as trouble reading and understanding the text, due in part to my handwriting, as well as the poor positioning of information.



Log Expense

For the scope of my app, I wanted to primarily focus my energy towards perfecting the experience for saving money, not spending money. My users don't typically deal with cash expenses, and banking apps and websites are a helpful way to track expenses, automatically. Therefore, my app does not require this feature.



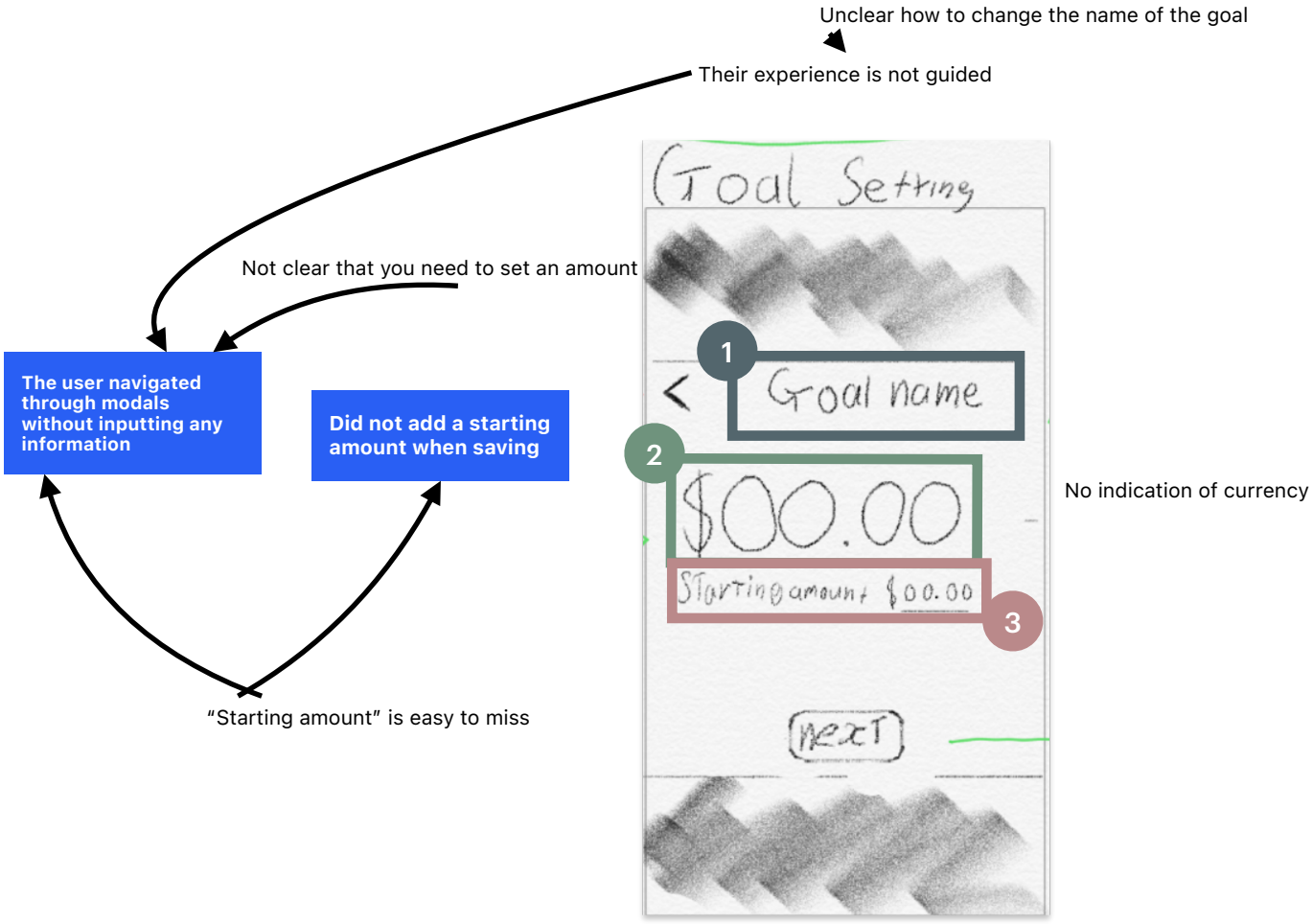
Outside the updated scope

Set Goal

One of the biggest problems with my paper prototype was the compression of information on these pages. The problem compounds when everything is hand drawn; it gets harder to distinguish elements from each other.

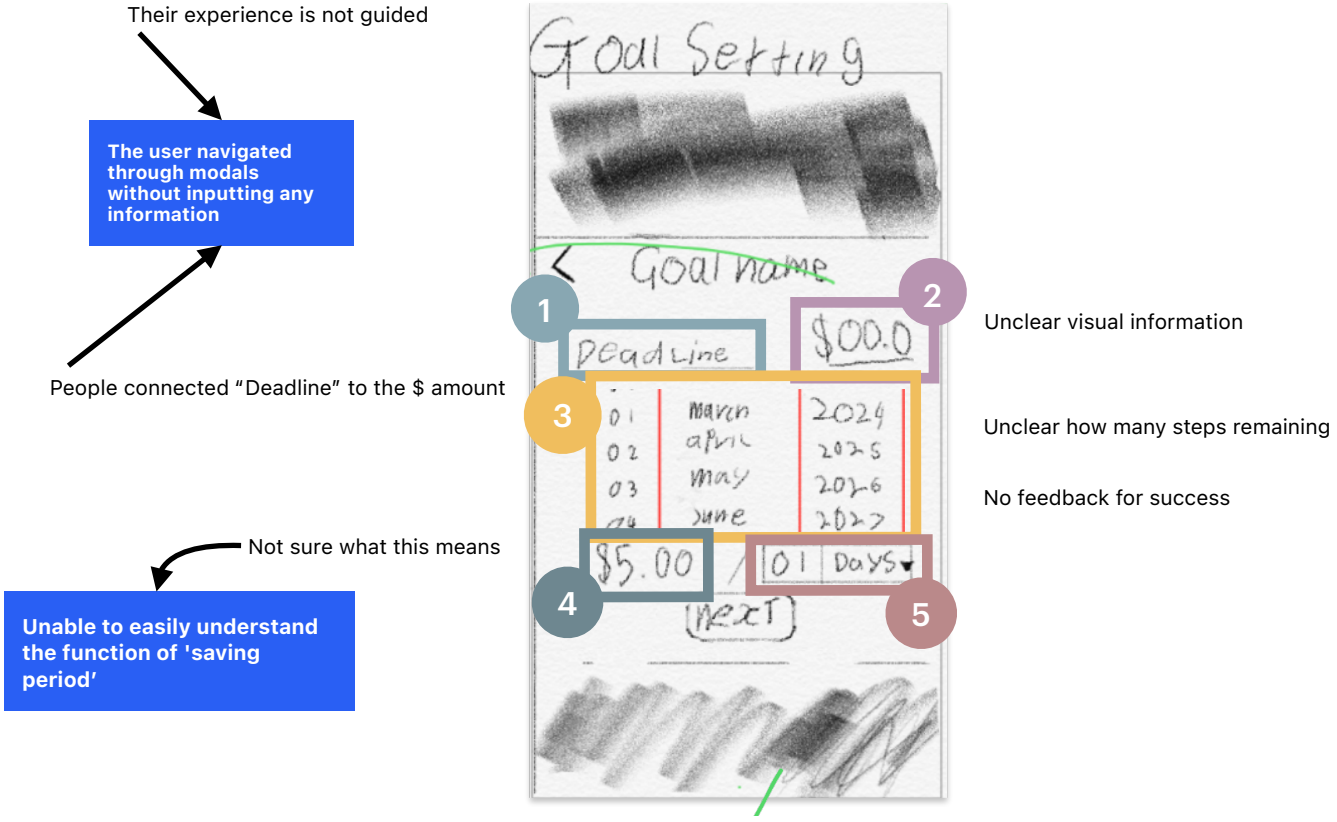
Screen 1.

This screen presented the user with too much information all at once, without offering users an opportunity to begin to break it down into smaller steps. On this screen, the user is expected to set the goal name **(1)**, the goal amount **(2)**, and the starting amount **(3)**, all in one place, with only the goal amount **(2)** standing out, albeit without clear indication of what it is or does.



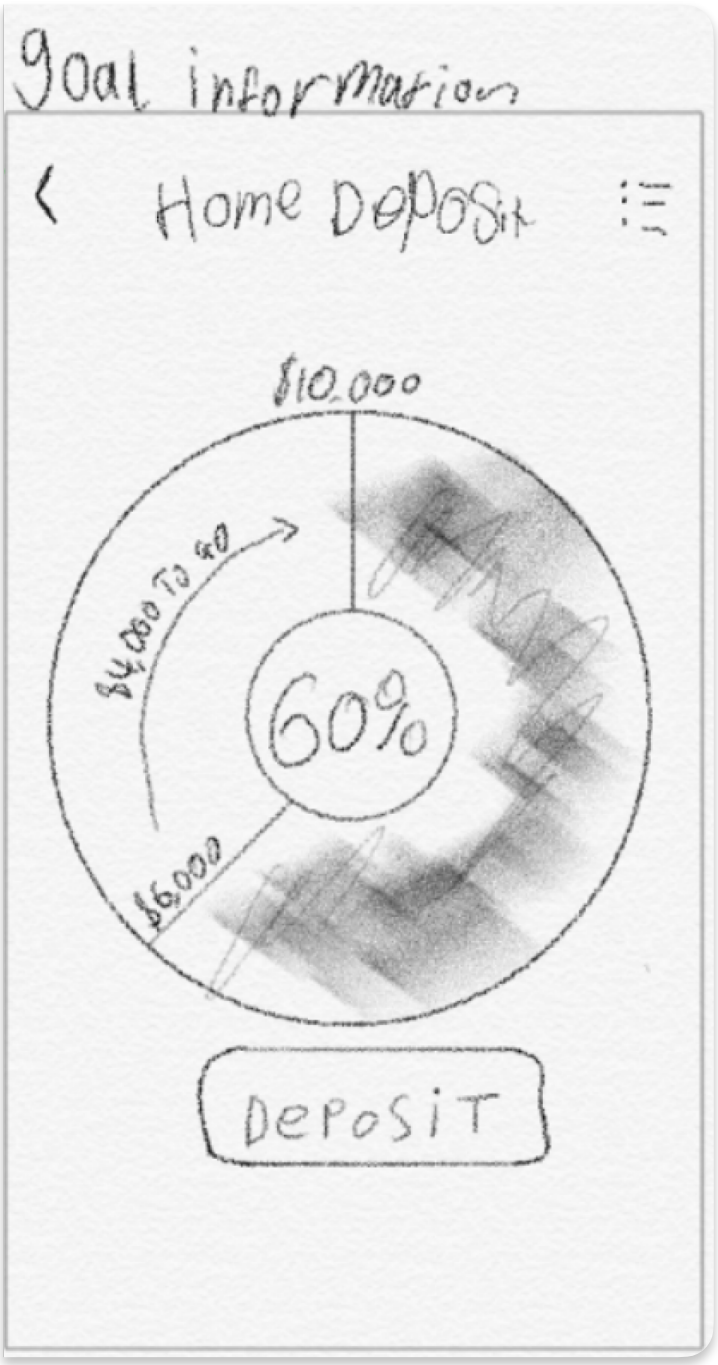
Screen 2

This screen presents more conflicting information to the user, with no clear way to understand what anything really means or does, or how information joins or separates with other information, for example: although this information was not intended to be connected, and users would have trouble identifying what each element represented, users would connect Deadline **(1)** to the Amount **(2)**, rather than to the date scroller **(3)**, because of left-to-right reading patterns, and poor grouping with Deadline**(1)** and the scroller**(3)**. Users also had trouble understanding how $5.00**(4)** connected to saving period information**(5)**, let alone what each element represented.



Goal Information

Upon reviewing my test results from A1, I realized that this screen was never intended to be interacted with, nor had any leading interactions from it. There were no problems with this screen, however it did not serve a purpose for the project requirements or the scope of testing my app's function, which was to set a saving goal.



Outside the updated scope

Brainstorm

Looking at my old wire flows, I could still find valuable insights. They offer me a chance to look at them and break them down into their parts, in order to see what my app really needs in order to achieve its purpose.

And the app's purpose was also due to change; I was able to look back at my wire flows and consolidate my app to just one primary function, and omit the secondary function, which could not solve it's desired problem, as the problem does not exist within the generation of people who would use this app.

I also updated the test scenario while I was brainstorming features, since the old scenario featured information that would be irrelevant for the current design.

- Allow any term of goal without restriction
- Integrate bank accounts, calculate spending patterns
- Visualize a photo to show progress towards the goal
  - Show a photo representation of the item goal in the goal overview on the home screen
  - Show the progress as a variable fill for the goal on the Home Screen
- Allow for creation of a new goal within just a few minutes
  - New goal from home screen
  - The progression through creating the goal should allow for spontaneous and quick train of thought
  - Have presets for common goals
    - Presets are just a starting point, but it can be useful to have which makes starting less daunting
- Provide tips on the homepage based on the progress of their goals
- Celebrate once the user has past certain milestones
- Show if the user is ahead / on track / behind on the Home Screen user is ahead / on track / behind on the home screen
- Outside the scope of my design
- Allow users start a goal off the home screen

Old:

1. You Just bought $60 worth of groceries with cash at queen vic market, and want to keep track of your spendings.
2. you needed to save up some money for a coffee machine for your friend's birthday present.

Information you need to know:
You are paid every 2 weeks
You need $1,500 by the end of June, before their birthday
You have $500 worth of spare money you can put towards this goal right now.
Calculate how much you need to save per paycheck

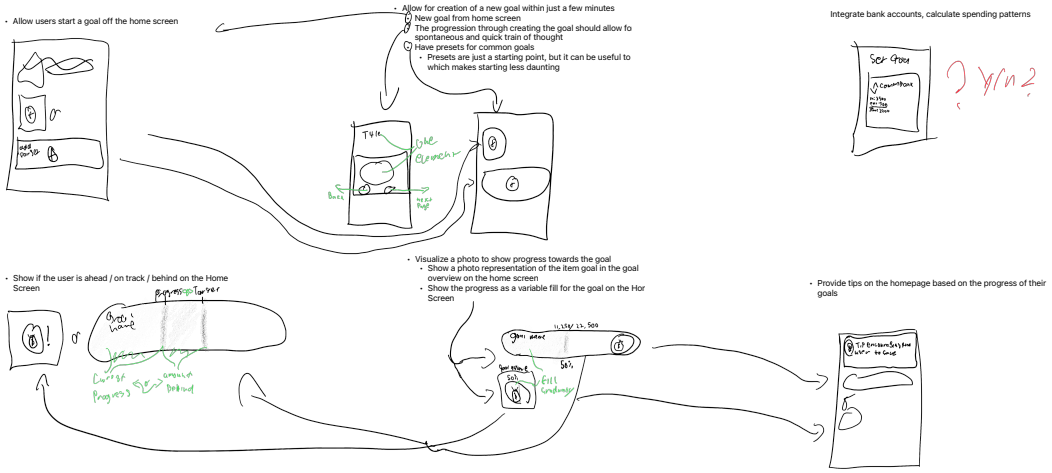
New:

you need to save up some money for a coffe machine for your friend's birthday present.
information you need to know:
You need $1,500 by the end of June 2025, before their birthday
Calculate how much you need to save per paycheck

Much simpler, and easier for the user to remember, without needless information.

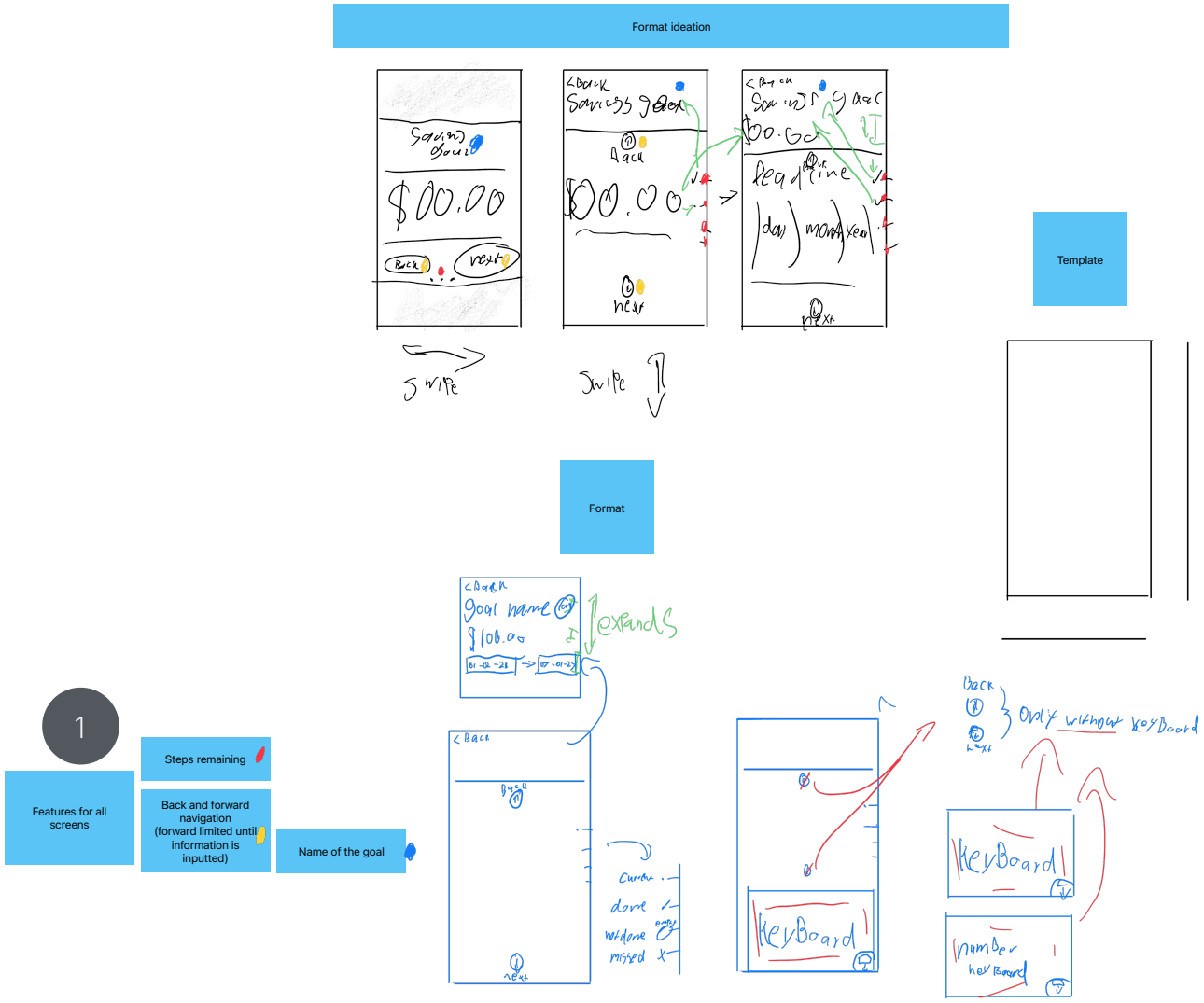
Consolidate Brainstorm and Visual Ideation

I took the information I gathered while brainstorming my app's features, and took them to the drawing board to iterate on them. I started with the visual idea of the old design in my head, but forced myself to draw things that weren't present in the original design.



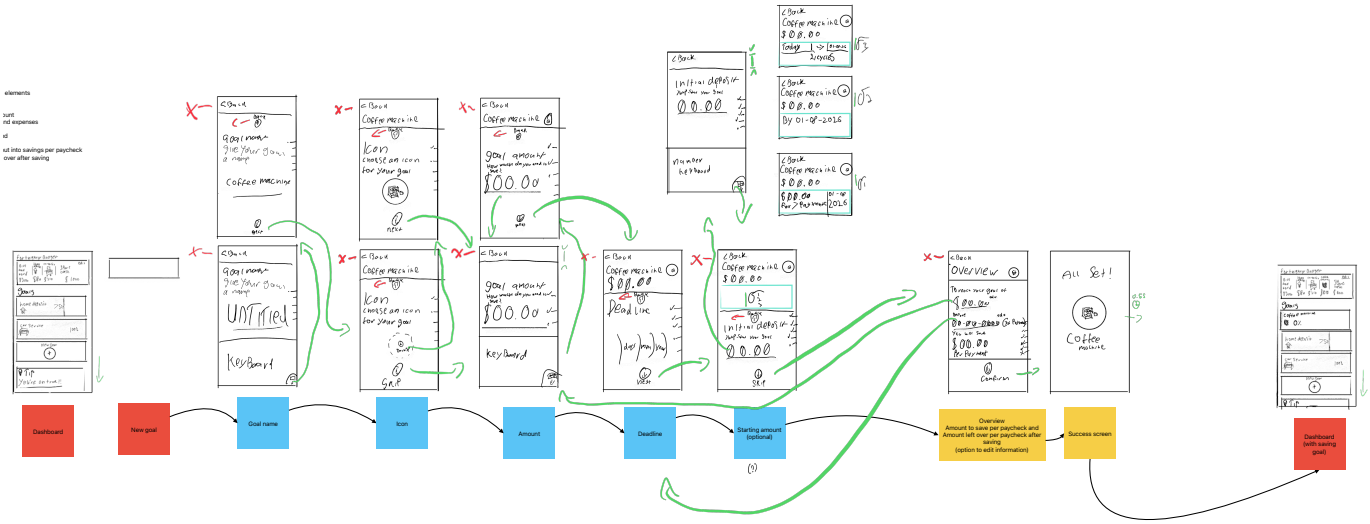
From this visual ideation, I realized that a lot of these functions are able to be put within single elements, they didn't have to be separate. As an extension of the idea of a 'single element', I chose to treat the entire process of saving as a 'single element', only with variations between screens as you progress. This idea that each stage in a design is just a version of the previous stage was essentially my own way of connecting each step together more fluidly. I began drawing ideas the process as a single screen in a medium fidelity.

I started with identifying features(1) that this 'single element flow' would have, not including the guts of each screen. From here, I sketched out an improved version of the paper wireframe, with features that it lacked, but disagreed with the blank space above and below. This vertical mindset drove me to the idea of having everything progress downwards vertically, building up as a block of information in the header, that builds in detail with the progression of the goal. To navigate with this navigation style, I decided to use vertical gestures, as well as buttons, and a line bar next to the guts, which showed the user what stage they were on, and how long they had to go.



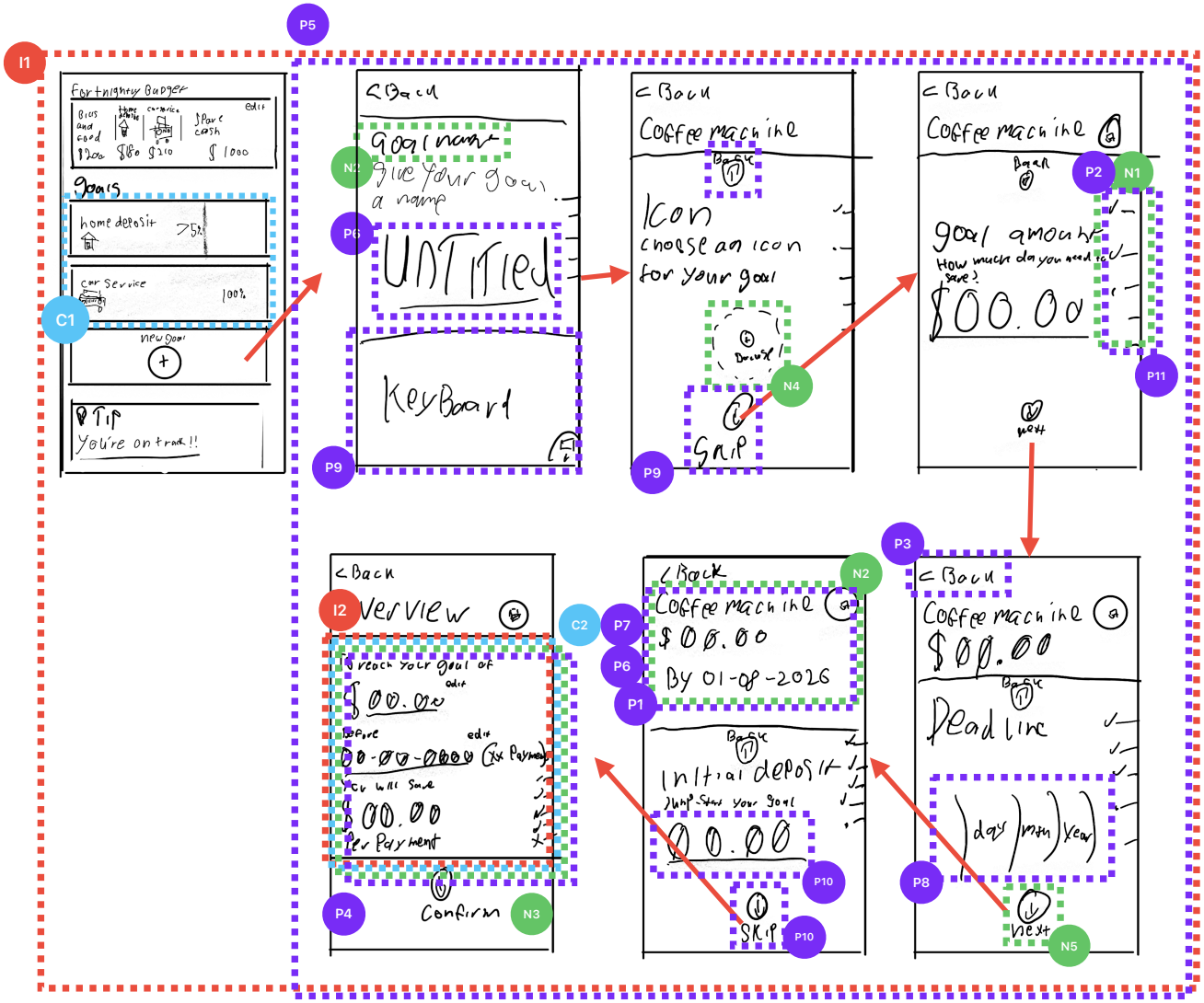
From here, I drew up slightly higher fidelity versions as a template to use, and considered what would happen when a keyboard was present. I decided that the keyboard as well as the buttons would take up too much space on the screen, so when the keyboard was present, navigation was hidden.

Improved Wire-flows



From the ideation stage came the wire-flows. These featured some elements that were present in the previous wire-flows, however they had a lot more room to breathe. I split the functions of the app into color-coded notes groups, gave a name for each screen based on the functions that I found when assessing my previous wireframes, and layered 'variations' (Keyboard, updated information, etc) of the same screen vertically.

3. Documentation (Annotation)



After designing and sketching out wire-flows, I digitally annotated elements based on these categories:

Information seeking behaviors (I#)

Categorization (C#)

Navigational models (N#)

UI Patterns (P#)

I named these annotation bubbles in a letter-number format rather than just numbered, in order to be able to add annotations to any category and retain a nice structure. I color coded them in order to differentiate them from each other further than just the lettering.

Annotation

Information Seeking Behavior

- I1 Known Item Seeking
- I2 Re-Finding

Categorizing

- C1 Chronology
- C2 Chronology

Navigating

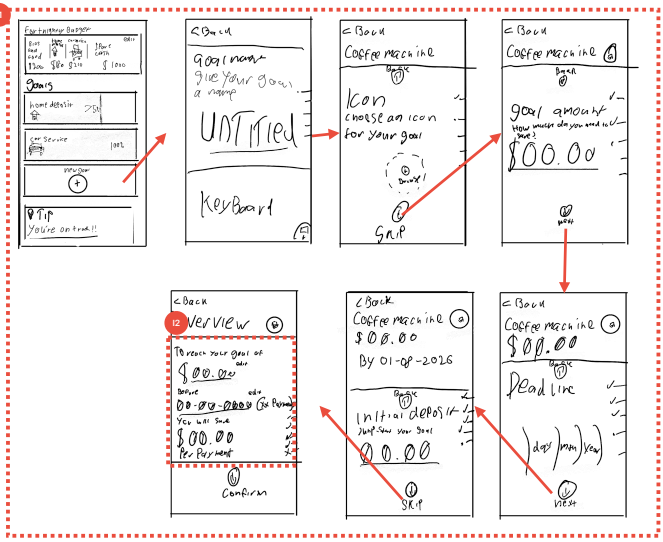
- N1 Help the user remember
- N2 Help the user remember
- N3 Contextual
- N4 Literal Actions
- N5 Literal Actions

UI Patterns

- P1 Progressive Disclosure
- P2 Pagination
- P3 Home Link
- P4 Article List
- P5 Continuous Scrolling
- P6 Fill in the Blanks
- P7 Input Feedback
- P8 Calendar Picker
- P9 Morphing controls
- P10 Good Defaults
- P11 Steps Left

I felt that just assigning a word to each element felt too ambiguous, and the big cloud of multicolored bubbles and squares was hard to follow, so I explain my reasoning in more detail below.

Information Seeking Behavior



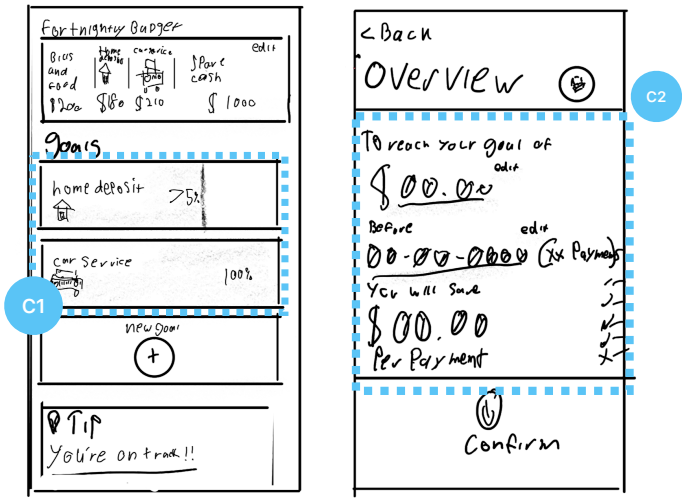
I1 - Known item seeking

The user will progress through a series of screens, the answers to which are (hopefully) pre-planned

I2 - Re-Finding

Probably a stretch, but the user is presented with a screen to return to any section of the flow before confirmation.

Categorization



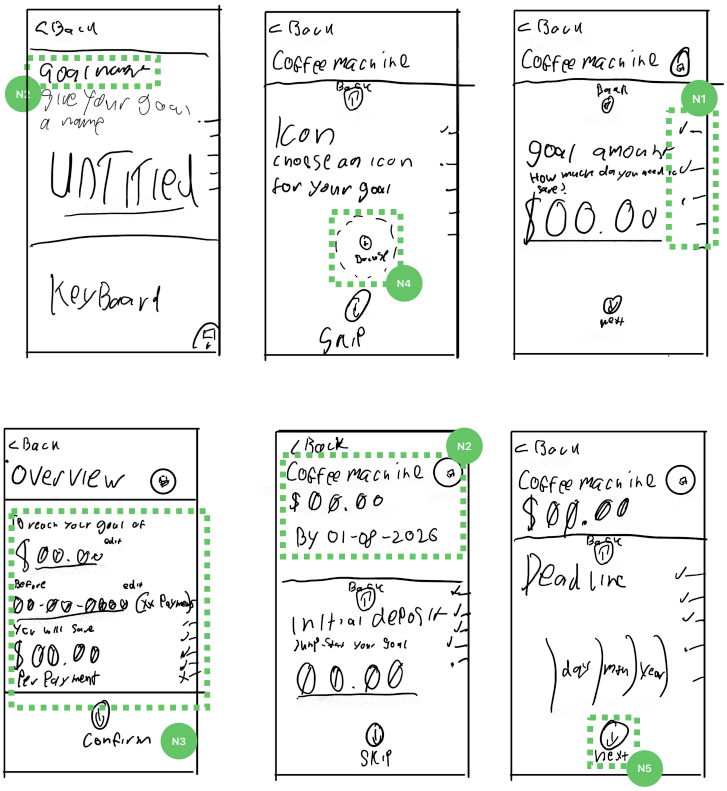
C1 - Chronology

Tabs are sorted by recency, with the most recent on top

C2 - Chronology

Differentiated from C1, the user is presented with information that is sorted from the top (first interaction) to bottom (most recent interaction) with an additional layer underneath that was not accessed until the point of this screen.

Navigation



N1 - Help the user remember 1

The user is shown a list of lines with an icon to represent what they have done on that screen (tick=passed, cross=skipped, blank=not seen yet, dot=current page).

N2 - Help the user remember 2

The user is shown a list of the result of their previous inputs to remind them, even subconsciously, what their goal was or is, and gives them the option to correct their mistakes.

N3 - Contextual

The user is shown a final overview of their goal, and can click on any of the previous inputs in order to change them. This screen is only shown in the **context** that all inputs are already complete.

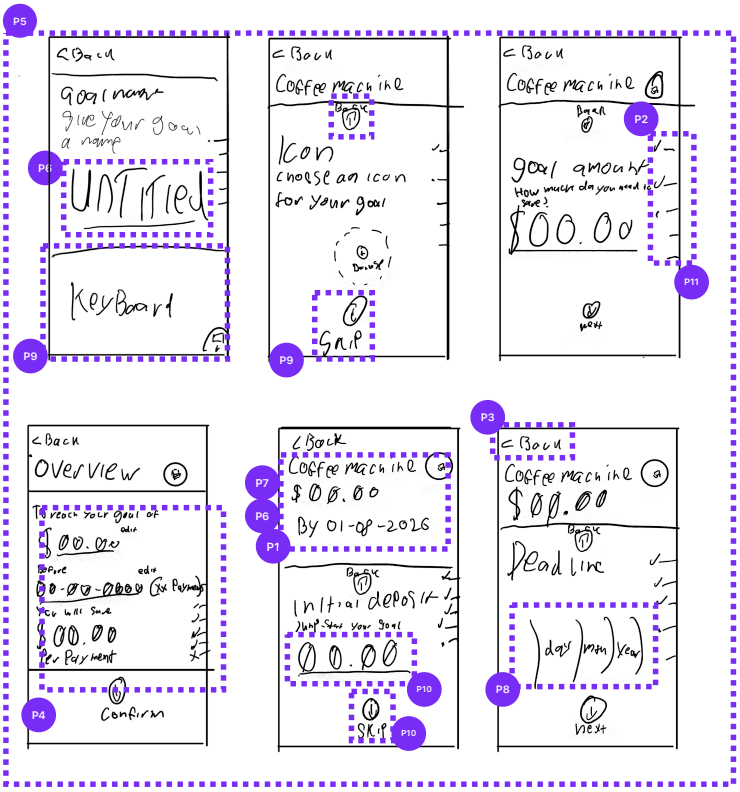
N4 - Literal Actions 1

'browse' is an action that the user can complete by looking at the selection of pictures.

N5 - Literal Actions 2

Maybe a stretch, but the user can navigate to the 'next' screen in the sequence

UI Patterns



P1 - Progressive Disclosure

The menu will grow as information is added to it, which is progressive as opposed to a static height equivalent, which would be static.

P2 - Pagination

Each control is on a different screen, or page.

P3 - Home Link

The user may return 'home' at any time

P4 - Article List

The user is presented with a list containing components of previous screens.

P5 - Continuous scrolling

The user flow in this design is progressed through scrolling to set-points in the vertical design.

P6 - Fill in the Blanks

Keywords like 'untitled' are there to represent information-less elements.

P7 - Input Feedback

The user is immediately presented with an update, showing them that the information they have inputted into the app has registered, as soon as they go to the next screen.

P8 - Calendar picker (https://ui-patterns.com/patterns/CalendarPicker)

It's a calendar of sorts, albeit scrollable, it is more common now in place of calendars.

P9 - Morphing controls

Controls move based on if there is screen space (keyboard present) or not

P10 - Good Defaults

On the 'initial deposit screen', the user may not have anything to put toward the goal immediately, so the default is 0 and the button is labelled as 'skip'

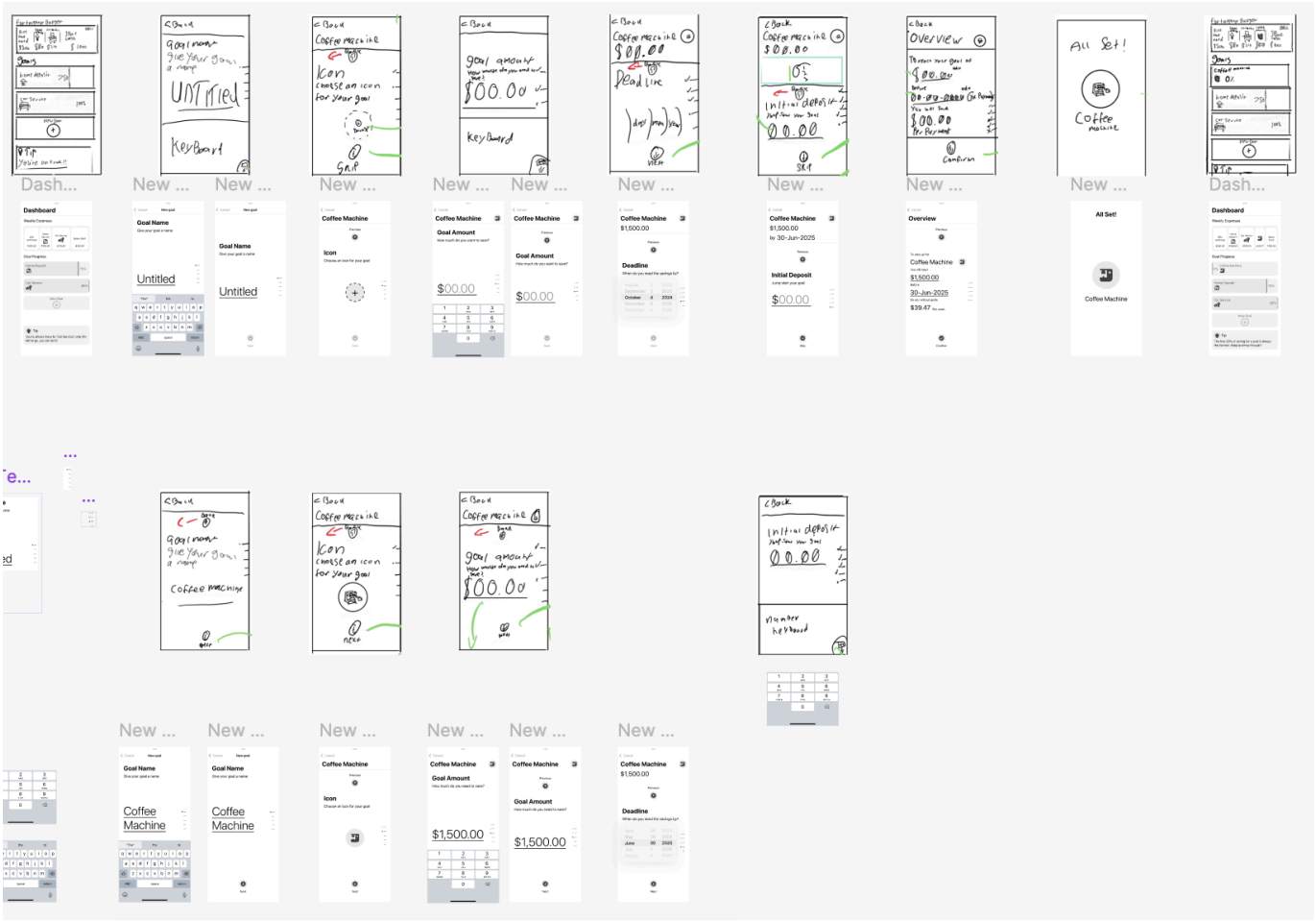
P11 - Steps Left

The user is shown their progress through the steps in the process of setting their savings goal.

4. High Fidelity Figma Prototype and Testing

Figma prototype

I took my high fidelity sketches (that we broke down in the last section) into Figma, and created a high fidelity digital prototype mockups with interactions and differentiations of each key state change. These are left uncolored by design, since this is a UX focused exercise, not a UI focused exercise.. yet.



https://www.figma.com/design/MroQlMv2F2rH9TdkELAAbF/Savings-App?node-id=114-846&t=WOaKoleNjYp14Bg8-1 **(page 2)**

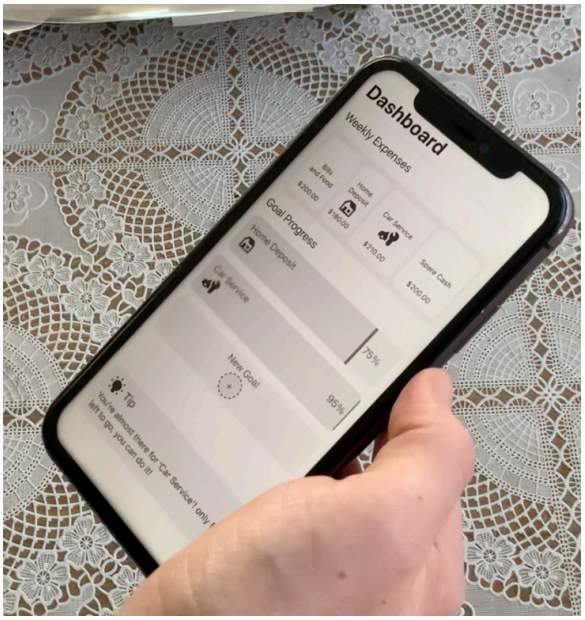
I made good use of components, variables and auto-layout in order to streamline the flow and make sure that everything was aligned and flows smoothly. This means that in the UI phase, I will be able to change the visual look of any element, and any related element will maintain consistency.

I used the Icons8 icon library, and the iOS design library, because those are what users are used to seeing and interact with. I also used a template for safe zones on the iPhone, which comes with layout guidelines and spacing guides for the sides of the screen.

I was able to disassemble and recreate the iOS Date picker based on the sample provided by the iOS design library, in order to give visual feedback to the user for interactions.

Testing

I was able to perform some user testing on my brother from home.



Scenario

you need to save up some money for a coffe machine for your friend's birthday present.

information you need to know:

You need $1,500 by the end of June 2025, before their birthday

Calculate how much you need to save per paycheck

https://rmiteduau-my.sharepoint.com/:v:/g/personal/s3972407_student_rmit_edu_au/ERif4QC70ANBt8NfRJ23cjABR8IVA-SKAipOl4bTOpOlsA?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=8onS5B

Post test questions

What did you like?

Straight forward design

Everything was presented well (the buttons tell you what they do)

What didn't you like?

Didn't like the color, too bland

The navigation interface isn't descriptive of the screen they represent

What did / didn't you understand?

Reported that they understood mostly all of it

Subconsciously, the user went to tap 'goal name' to change the name of the goal.

What would you have designed differently?

Would have made the navigation interface interactable and make them better represent the page

Did anything happen outside of your expectations, Or need explanation?

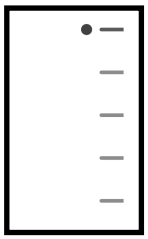
Although not reported by the user, their interactions with the prototype when setting the savings goal amount were misinterpreted by the app, causing them to go back to the first screen in the process.

The user initially attempted to scroll to change the date in the deadline view, which didn't work.

Problems with my current design

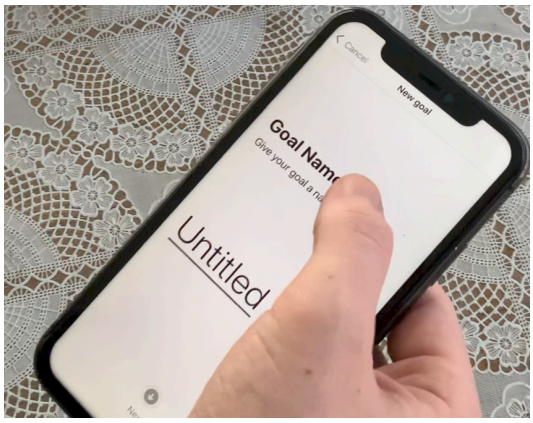
Excluding UI centered problems like color, I want to explore what I would improve about my app if I was to develop it further, or to address and attempt to understand issues that arose within user testing, in order to better understand what users expect in this app.

1. The navigation interface isn't descriptive of the screen they represent



This is in reference to the sidebar navigation, with the checks, crosses and dots, to show the user which screen they are on and how much they have left to go. The user suggested to have some sort of visual indication of what they represent. Something like an icon or a name to the line would help for navigation.

2. User subconsciously tapped 'goal name'



On the first screen in the goal-setting process, where the user was instructed to set a goal, they instinctively tapped on the text that read 'goal name'. I believe this is due to any combination of these two:

a. The user perceives the text more like a button than the text field.

b. It is not obvious when the text field is selected, regardless of if the keyboard is present.

The solution to these problems might be as straightforward as placing a flashing | cursor in place of the text field, as an indication that the user is typing in that field. The solution might also include thinning the weight of the 'goal name' text.